

Multiple Choice (10 marks)

1. Which of the following characteristics is predicted for an early-successional plant community?
 - (a) High detrital biomass
 - (b) High number of predatory species
 - (c) High presence of fully matured plant species
 - (d) High frequency of R-selected species
 - (e) High rates of soil nutrient depletion
2. In what way does Lamarck's theory of acquired characteristics not agree with present evidence?
 - (a) Acquired characteristics are inherited through a blending of parental traits
 - (b) Acquired characteristics are inherited but do not influence evolution
 - (c) Acquired characteristics are inherited randomly, regardless of their use
 - (d) Acquired characteristics cannot be inherited
 - (e) Acquired characteristics are inherited based on use and disuse
3. Why are organ transplants generally unsuccessful?
 - (a) Lack of available organs
 - (b) Graft rejection due to immunological reaction
 - (c) Incompatibility of blood types
 - (d) Allergic reactions to anesthesia
 - (e) Insufficient surgical expertise
4. How does the body control the amount of water excreted?
 - (a) The body controls water excretion through sweat glands.
 - (b) Water excretion is determined by the concentration of electrolytes in the body's cells.
 - (c) The kidneys control water excretion by changing the size of the glomerulus.
 - (d) Water excreted is controlled by the hormone insulin.
 - (e) The body adjusts water excretion by altering the pH level of the blood.
5. During which phase of the cell cycle does the quantity of DNA in a eukaryotic cell typically double?

- (a) Telophase
- (b) G₂
- (c) Prophase
- (d) Interphase
- (e) Metaphase

True / False (10 marks)

Answer True or False

6. True or False: Does a 4 diagram manual enable laypersons to operate the Laryngeal Mask Supreme®?
7. True or False: Is laparoscopic adrenalectomy safe and effective for adrenal masses larger than 7 cm?
8. True or False: Are physicians aware of the side effects of angiotensin-converting enzyme inhibitors?
9. True or False: c-Kit-dependent growth of uveal melanoma cells: a potential therapeutic target?
10. True or False: Could ADMA levels in young adults born preterm predict an early endothelial dysfunction?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the physiological reasons for the differences in blood pressure readings taken from the upper arm versus the leg, considering the impact of distance from the heart on blood pressure.
12. Discuss the scientific beliefs about Earth's age and the constancy of species prior to Lyell and Darwin, and analyze how these views influenced their theories of evolution.

End of Paper